COD-PS Nigeria
Update Year 2023 / Reference Year 2022

Objective of This Explanatory Technical Note

This explanatory technical note provides supplementary notes on the strengths and limitations of each Common Operational Dataset on Population Statistics (COD-PS) against the projections by the WPP 2022 Revision of the UN to enable informed humanitarian decision-making.

Structure of Supplementary Note

This supplementary note is organized into the following sections:
1. Metadata
2. Methodological Documentation
3. Intrinsic Population Growth Rates (ADM-0 & 1)
4. Population counts by age and sex (ADM-0)
5. Relative population size by age and sex (ADM-1)

1. Metadata

Item	Metadata
Country	Nigeria
Source(s)	National Population Commission, Nigeria
Source(s) Link(s)	https://nationalpopulation.gov.ng/publications
Population Data Type (Census enumeration, Direct estimate, or Population projection)	Population projection
Baseline Population Year	2006
COD-PS Reference Year	2022
COD-PS Publication Year	2023
ADM-1 Name	State
ADM-1 Number of Units	37
ADM-1 Sex and Age Disaggregation	Sex and age disaggregated by 5-year age groups
ADM-1 Open-ended Age Group	80+
ADM-2 Name	N/A

Item	Metadata
ADM-2 Number of Units	N/A
ADM-2 Sex and Age Disaggregation	N/A
ADM-2 Open-ended Age Group	N/A
ADM-3 Name	N/A
ADM-3 Number of Units	N/A
ADM-3 Sex and Age Disaggregation	N/A
ADM-3 Open-ended Age Group	N/A
General Notes	
Data Limitations	
COD-PS Demographic Data Quality Assessment	The COD-PS was assessed against ADM-0 estimates and projections published by the World Population Prospects (2022 Revision), and found to be generally consistent at the ADM-0 and ADM-1 level.

2. Methodological Documentation

Item	Methodological Documentation
Methodology Used	This Common Operational Dataset on Population Statistics (COD-PS) is estimated using cohort-component projections developed by the National Population Commission, using the 2006 census as a baseline.
Baseline Population	2006 census
Post-enumeration survey (PES)	A post-enumeration survey was carried out after the 2006 census, although the PES report could not be located online.
Assessment and Adjustment of the Baseline Population	In order to account for age heaping, the baseline population was smoothed using the Arriaga Light method.
Fertility (births)	Fertility assumptions were based on data from the 2008, 2013, and 2018 Nigeria Demographic and Health Surveys, interpolating between years with available data. Projections of future fertility were based on assessment of the progress of the demographic transition in Nigeria and past fertility trends.
Mortality (deaths)	Mortality assumptions were based on historical data from sources such as the 2016 Human Development Index report by the National Bureau of Statistics, with an assumption of a steady increase in life expectancy by sex. Age patterns of mortality were based on the Coale-Demeny North lifetable.
International migration (net migration)	

Item	Methodological Documentation
Internal migration (migration within country)	Assumptions about internal migration incorporated data from the 2006 census as well as a 2010 survey on migration, with migration being defined as an individual having changed their local government area (LGA) of residence for at least 6 months in the past 10 years. Information on the age-sex patterns of migration were based on model schedules of gross migration developed by Castro and Rogers (United Nations, 1989). Net migration was assumed to remain constant throughout the projection period.

3. Intrinsic Population Growth Rates (ADM-0 & 1)

At ADM-0, the implied post-censal (2006-2022) population growth rates of this COD-PS (2.8% per year: female: 2.91%, male: 2.69%) are higher than the UN's population projections published in the 2022 revision of the World Population Prospects (WPP) (2012: 2.73%, 2016: 2.52%, 2019: 2.44%). This is likely shaped in differences in projection assumptions and methods used.

Since the last census, this COD-PS projects the highest growth rates for the states of Zamfara (8.9%), Federal Capital Territory (4.88%), and Katsina (3.63). In contrast, the three states with the lowest projected growth are Akwa Ibom (1.52%), Osun (1.63%), and Kogi (1.86%). These projected growth rates reflect the projection assumptions used, especially related to migration and fertility.

ADM-1	Last Census, Female	Last Census, Male	CODPS, Female	CODPS, Male	PGR(%), Female	PGR(%), Male	PGR(%), Both
Abia	1,430,298	1,415,082	2,094,668	2,048,425	2.45	2.24	2.35
Adamawa	1,607,270	1,571,680	2,446,708	2,455,347	2.77	2.65	2.71
Akwa Ibom	1,983,202	1,918,849	2,487,982	2,491,436	1.62	1.43	1.52
Anambra	2,117,984	2,059,844	2,983,151	2,970,366	2.31	2.11	2.21
Bauchi	2,369,266	2,283,800	4,160,248	4,148,535	3.75	3.50	3.62
Bayelsa	874,083	830,432	1,267,098	1,270,277	2.64	2.34	2.49
Benue	2,144,043	2,109,598	3,105,106	3,036,178	2.42	2.17	2.30
Borno	2,163,358	2,007,746	3,029,395	3,082,067	2.57	2.21	2.39
Cross River	1,471,967	1,421,021	2,207,001	2,199,203	2.75	2.51	2.63
Delta	2,069,309	2,043,136	2,829,285	2,806,860	2.03	1.91	1.97
Ebonyi	1,064,156	1,112,791	1,703,744	1,538,774	2.66	2.31	2.49
Edo	1,633,946	1,599,420	2,382,155	2,394,887	2.49	2.39	2.44
Ekiti	1,215,487	1,183,470	1,799,478	1,792,685	2.62	2.43	2.53
Enugu	1,596,042	1,671,795	2,451,113	2,238,940	2.39	2.12	2.26
Federal Capital	733,172	673,067	1,484,198	1,583,259	4.94	4.81	4.88

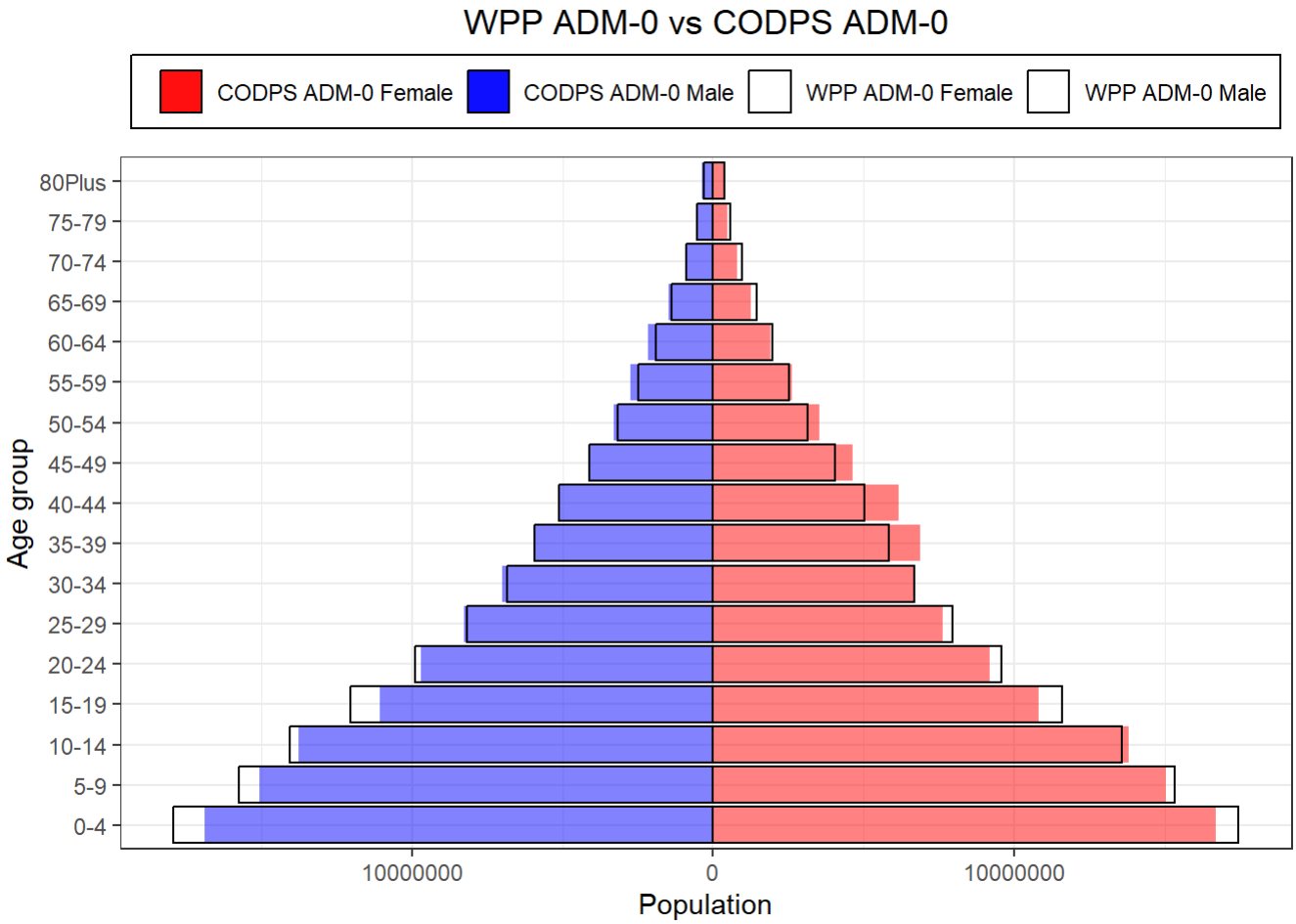
ADM-1	Last Census, Female	Last Census, Male	CODPS, Female	CODPS, Male	PGR(%), Female	PGR(%), Male	PGR(%), Both
Territory							
Gombe	1,244,228	1,120,812	1,907,881	2,052,241	3.32	3.13	3.22
Imo	1,976,471	1,951,092	2,748,610	2,710,727	2.14	1.97	2.06
Jigawa	2,198,076	2,162,926	3,818,803	3,704,594	3.55	3.26	3.40
Kaduna	3,090,438	3,023,065	4,581,564	4,450,617	2.60	2.28	2.44
Kano	4,947,952	4,453,336	7,436,564	8,025,613	3.20	3.02	3.11
Katsina	2,948,279	2,853,305	5,203,404	5,165,079	3.76	3.50	3.63
Kebbi	1,631,629	1,624,912	2,829,460	2,734,447	3.47	3.23	3.35
Kogi	1,672,903	1,641,140	2,247,148	2,219,653	1.96	1.77	1.86
Kwara	1,193,783	1,171,570	1,791,950	1,759,073	2.66	2.42	2.54
Lagos	4,719,125	4,394,480	6,542,571	6,949,233	2.49	2.42	2.46
Niger	2,004,350	1,950,422	3,426,921	3,356,404	3.52	3.22	3.37
Ogun	1,864,907	1,886,233	3,232,585	3,146,947	3.37	3.27	3.32
Ondo	1,745,057	1,715,820	2,670,882	2,645,721	2.77	2.60	2.68
Osun	1,734,149	1,682,810	2,219,961	2,215,842	1.73	1.53	1.63
Oyo	2,802,432	2,778,462	4,020,046	3,956,035	2.31	2.15	2.23
Plateau	1,598,998	1,607,533	2,408,394	2,308,911	2.53	2.30	2.42
Rivers	2,673,026	2,525,690	3,685,567	3,791,238	2.36	2.18	2.27
Sokoto	1,863,713	1,838,963	3,237,546	3,153,501	3.54	3.29	3.42
Taraba	1,171,931	1,122,869	1,800,432	1,809,411	2.95	2.71	2.83
Yobe	1,205,034	1,116,305	1,794,832	1,854,775	2.97	2.70	2.84
Zamfara	733,172	673,067	2,959,893	2,873,601	9.26	8.54	8.90
Total	69,493,236	67,196,543	106,996,344	106,940,902	2.91	2.69	2.80

4. Population Counts by age and sex (ADM-0)

Below, we compare the population structure across sex and age groups between the CODPS and WPP projections at the national level.

The age- and sex-specific population counts of this COD-PS are largely consistent with the WPP 2022 Revision projections at ADM-0 level, with the NPC projections tending to estimate slightly lower numbers of infants and young children than the WPP projections, and slightly higher numbers of working-age women. One of the largest differences is seen for the 0-4 age interval, with a difference between the WPP and COD-PS projections of 4.31% for females and 5.75% for males.

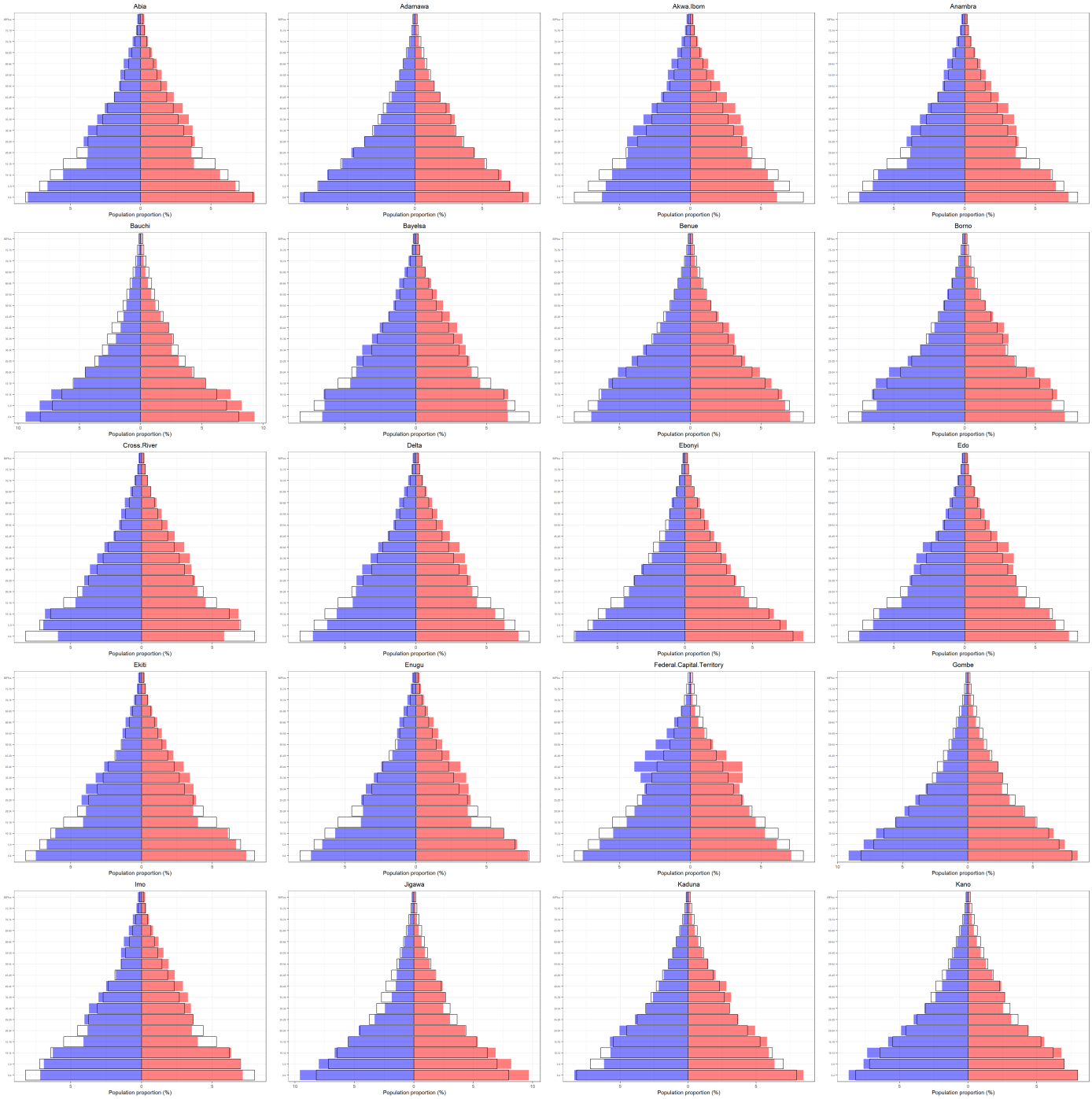
	CODPS-WPP, Female	Difference(%)	CODPS-WPP, Male	Difference(%)
0-4	-751,682	-4.31	-1,029,978	-5.75
5-9	-293,416	-1.92	-686,845	-4.37
10-14	223,067	1.64	-286,406	-2.04
15-19	-792,331	-6.84	-970,106	-8.06
20-24	-368,828	-3.86	-200,458	-2.02
25-29	-329,268	-4.14	79,668	0.97
30-34	57,018	0.86	183,714	2.69
35-39	1,029,869	17.65	35,486	0.60
40-44	1,121,758	22.25	-57,183	-1.12
45-49	573,774	14.16	-28,502	-0.69
50-54	367,981	11.68	159,732	5.07
55-59	95,114	3.77	281,330	11.36
60-64	-79,014	-3.99	255,872	13.48
65-69	-210,219	-14.38	94,150	6.89
70-74	-172,382	-17.89	45,178	5.09
75-79	-116,367	-19.85	-22,661	-4.21
80Plus	-15,180	-4.01	49,284	15.45

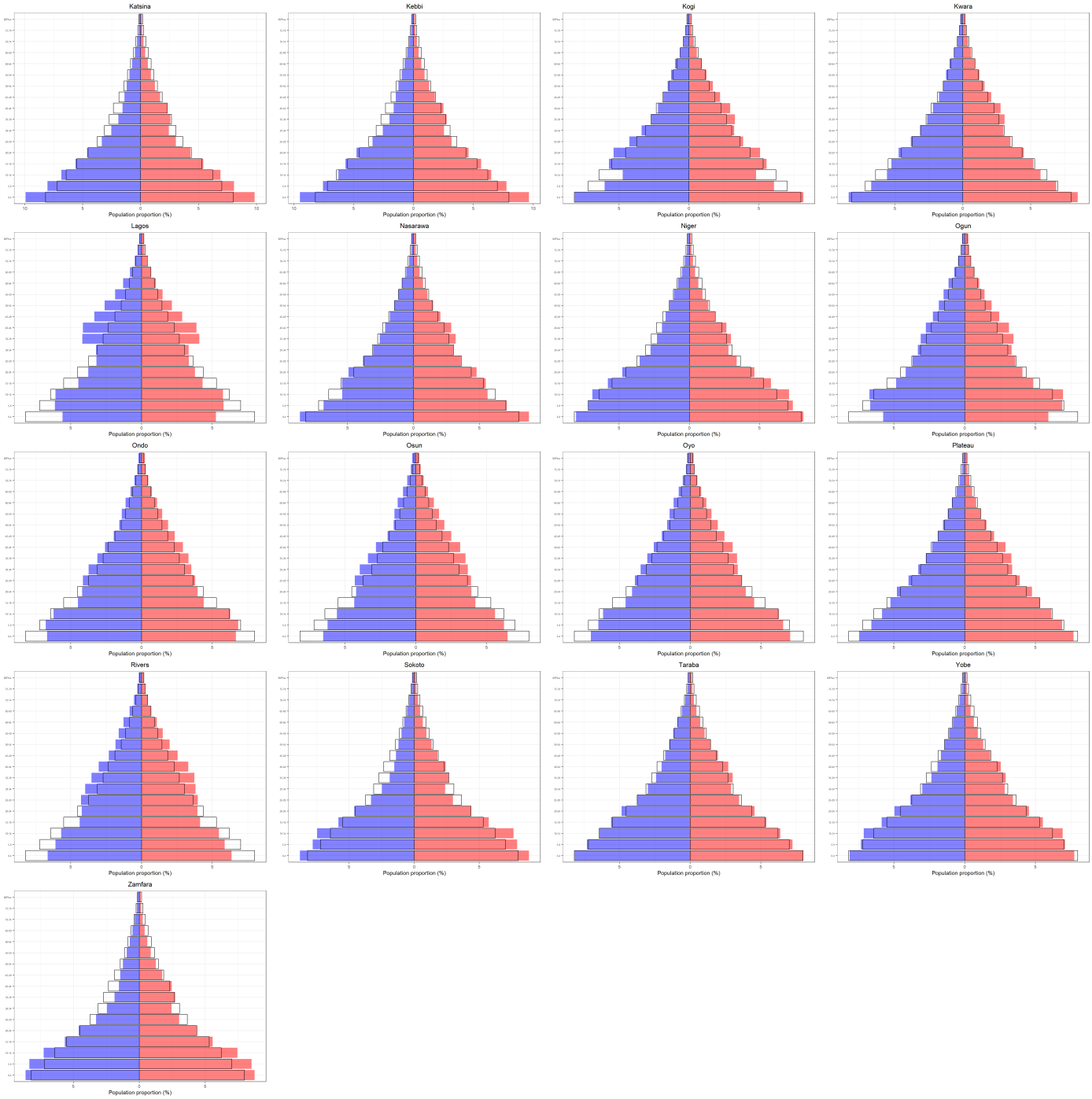


5. Relative population size by age and sex (ADM-1)

Below, we compare the relative population proportions for 5-year age groups by sex between the CODPS ADM-1 projections and the WPP ADM-0 projection.

While the sex and age proportions of many states generally follow the WPP ADM-0 projections, Lagos and the Federal Capital Territory both have significantly higher shares of working-age adults, and a lower share of children and young adults than the WPP projections. This likely reflects labor migration at working ages for economic opportunities associated with urban economic centers.


Abia
Adamawa
Akwa.Ibom
Anambra
Bauchi
Bayelsa
Benue
Borno
Cross.River
Delta
Ebonyi
Edo
Ekiti
Enugu
Federal.Capital.Territory
Gombe
Imo
Jigawa
Kaduna
Kano
Population proportion (%)


Katsina
Kebbi
Kogi
Kwara
Lagos
Nasarawa
Niger
Ogun
Ondo
Osun
Oyo
Plateau
Rivers
Sokoto
Taraba
Yobe
Zamfara
Population proportion (%)